

Decadal mapping of agave cultivation across the three core mezcal-producing districts of Oaxaca (Tlacolula, Yautepec, Miahuatlán; 2017–2025) using AlphaEarth Foundations, terrain stacks, and a histogram-gradient-boosting classifier with a data-driven decision threshold

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May 18, 2026

Abstract

Context. Mezcal exports from Mexico grew roughly twentyfold between 2015 and 2024, concentrating pressure on *Agave* cultivation in Oaxaca, which produces over 85% of certified mezcal. Despite intense ethnographic interest in the resulting land-use change, no peer-reviewed study has quantified annual agave extent at 10 m resolution across the boom corridor for the entire 2017–2025 window.

Methods. We construct a per-pixel, annual binary classifier of *Agave* cultivation across the three core mezcal-producing INEGI districts of Oaxaca – **Tlacolula** (25 municipios), **Yautepec** (12) and **Miahuatlán** (32) – comprising 69 municipios within a 1.53° E–W $\times$ 1.22° N–S bounding box and an actual muni-union land area of $\sim 11,850$ km². Predictions outside the 69-muni union are explicitly masked out so that all area estimates reflect land actually administered by one of the three districts. Predictors stack the 64 dimensions of Google’s *AlphaEarth Foundations* annual embeddings with five topographic descriptors (elevation, slope, northness, eastness, and the Terrain Ruggedness Index) derived from Copernicus DEM GLO-30. A histogram-based gradient boosting classifier (HGB, `class_weight='balanced'`) is trained on 4,849 manually photo-interpreted reference polygons (500 unique 80 m centroids $\times$ multi-year, 2016–2025; collected by the iaor labelling team). The decision threshold is set data-driven at 0.60, selected as the global F1 optimum on the held-out test panel (per-year F1 sweep in 0.05 increments). Annual predictions are reported as raw map area with 95% bootstrap confidence intervals together with a commission-only correction (raw $\times$ user’s accuracy) following [Olofsson et al. \(2014\)](#).

Results. At the production threshold 0.60, the classifier attains overall accuracy 0.9425, Cohen’s $\kappa = 0.8018$, recall 0.821 and precision 0.853 for the agave class on an 870-point held-out panel (per-class confusion: TN = 692, FP = 22, FN = 28, TP = 128). Across the three-district AOI, raw agave extent grows from 4,289 ha (95% CI 4,263–4,314) in 2017 to 18,694 ha (95% CI 18,642–18,746) in 2025, a net **4.4-fold increase (+336%)** over the eight-year window; the commission-corrected estimator (raw $\times$ precision 0.853) gives the same percentage growth from 3,658 ha to 15,946 ha. The trajectory is strongly non-monotonic. After an initial pulse in 2018 (+44%), agave area contracts through 2019–2020 (–5%, –22%), recovers modestly in 2021–2022 (+6%, +30%) and then enters a steep acceleration with the largest single-year expansion of the panel in 2024 (+107%, agave area doubling in twelve months) followed by a further +49% in 2025.

Conclusions. Across the three core mezcal-producing districts of Oaxaca (Tlacolula, Yautepec, Miahuatlán; 69 municipios, $\sim 11,850$ km²), the post-2017 agave boom is a ~ 4.4 -fold expansion in raw map area, a net gain of $\sim 14,400$ ha. The expansion is concentrated in a 2024–2025 acceleration that more than triples 2017 extent in two years and is the largest year-on-year change in the panel. This is the first decadal, per-pixel, area-adjusted record of agave cultivation restricted to the three administrative districts that anchor the Oaxaca mezcal denomination of origin at 10 m resolution. The methodology is fully open-source and reproducible on free public infrastructure.

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github.com/iortiz1891/agave-boom-oaxaca.

Code, data and interactive dashboard available at

Keywords: agave; mezcal; Oaxaca; land-cover change; AlphaEarth Foundations; histogram gradient boosting; biocultural regions; Olofsson area adjustment; commodity-driven agricultural expansion.

1 Introduction

The mezcal industry in Oaxaca has undergone an order-of-magnitude expansion since 2015. Certified mezcal exports from Mexico rose from roughly 1 million litres in 2015 to over 20 million in 2024 ([Consejo Regulador del Mezcal, 2024](#)), and Oaxaca’s share of national production is estimated at 85% or more. Ethnographic and case-study evidence from the Tlacolula sub-valley, Sola de Vega, and Miahuatlán documents intense conversion of milpa, fallow, scrubland and seasonally dry forest to *Agave* plantation ([Bowen, 2015](#); [Lira et al., 2022](#); [Robson et al., 2022](#)). The resulting land-use change has implications for water balance, soil erosion, biodiversity of wild and semi-wild agave populations, and the governance of the Mezcal Denomination of Origin ([Torres-García et al., 2019](#); [Vargas-Ponce et al., 2007](#); [Rangel-Landa et al., 2016](#)).

Despite this intense interest, no peer-reviewed study has quantified agave extent at fine spatial resolution and annual temporal cadence across the boom corridor for the entire post-2017 window. Producer-side records (the SIAP cierre agrícola and the Consejo Regulador del Mezcal padrón) reflect only certified producers and lag actual extent. Remote sensing of *Agave* is challenging because the canopy signal varies non-monotonically with stand age, is highly clustered in small parcels, and is embedded in a complex matrix of milpa, oak woodland, xeric scrub and degraded secondary vegetation.

Two recent developments unlock a tractable approach. First, the *AlphaEarth Foundations* collection ([Google, 2024](#)) provides pre-computed 64-dimensional annual embeddings of every 10 m pixel on Earth, derived from a Vision Transformer pre-trained on global Sentinel-2 imagery. The embeddings encode spectral, temporal and textural context without requiring per-task feature engineering. Second, the Copernicus GLO-30 digital elevation model supplies a complementary terrain stack that disambiguates plantation slope and aspect from spectrally similar cover types.

This paper presents the first decadal, per-pixel, area-adjusted record of agave cultivation across the three-district biocultural footprint of the Oaxaca mezcal denomination of origin for 2017–2025. We combine AlphaEarth embeddings with a five-band terrain stack and a histogram-based gradient boosting classifier, validate on an independent hold-out, and compute commission-corrected area with bootstrap confidence intervals.

2 Study area

The area of interest is the union of **69 municipios across three INEGI districts** that anchor mezcal production in Oaxaca: **Tlacolula** (25 municipios), **Yautepec** (12) and **Miahuatlán** (32). The combined footprint spans $[-97.0904^\circ, 15.9375^\circ]$ to $[-95.5560^\circ, 17.1550^\circ]$ (1.53° E–W, 1.22° N–S, ≈ 155 km by 135 km) with a muni-union land area of $\sim 11,850$ km². The membership list is fixed at the `cve_mun` level in `data/reference/v11_aoi_munis.csv`; predictions outside these municipios are explicitly masked out so that all area estimates reflect only land actually administered by one of the three districts.

The three districts span distinct biophysical and productive gradients within the Oaxaca mezcal denomination of origin. **Tlacolula** is the historic espadín productive core: a semi-arid (BSH–BSk) interior corridor at 1,500–2,000 m elevation on Calcisols, Vertisols and Leptosols, encompassing Santiago Matatlán, Tlacolula de Matamoros, San Pablo Villa de Mitla, San Lorenzo Albarradas and the Mitla piedmont. **Yautepec** extends east and south-east of the Tlacolula corridor across the upper Yautepec basin (Asunción Tlacolulita, San Carlos Yautepec, Nejapa de Madero, Santa María Ecatepec); it is rugged, sparsely populated, and an emerging production zone for both espadín and landrace species (*A. marmorata*). **Miahuatlán** runs south-west from the Tlacolula corridor across the Sierra Sur, ranging from the cabecera at Miahuatlán de Porfirio Díaz down toward the Pacific piedmont (San Mateo Río Hondo, San Miguel Suchixtepec, Santiago Xanica), and hosts a diverse mix of landrace species (*A. marmorata*, *A. karwinskii*, *A. potatorum*, *A. angustifolia*). District boundaries come straight from INEGI’s `cve_mun` attribute on `municipalities_real.geojson`; no hand-curation is applied.

3 Data

3.1 Predictors

AlphaEarth Foundations annual embeddings. The Google AlphaEarth collection (GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL, Google 2024) provides 64-dimensional embeddings per 10 m pixel and calendar year, available from 2017 onward. Each embedding compresses one year of Sentinel-2 observations through a Vision Transformer trained on a global self-supervised objective. We use the unmodified 64-band raster as the spectral predictor stack.

Terrain stack. From Copernicus DEM GLO-30 we derive five predictors at the AlphaEarth native grid: **elevation** (m), **slope** (degrees), **northness** (cos of aspect), **eastness** (sin of aspect), and the **Terrain Ruggedness Index** (TRI, Riley et al. 1999, the mean absolute elevation difference between a focal cell and its eight neighbours). The five terrain bands are constant in time and are stacked with the year-specific AlphaEarth embedding to form a 69-band predictor per pixel-year.

3.2 Reference data

The reference panel consists of **4,849 usable labels** obtained from 500 unique 80 m $\times$ 80 m polygons photo-interpreted across the 2016–2025 window (one label per polygon-year), produced by three labellers (iaor, iaor_local, iaor_2) on Sentinel-2 RGB imagery in a custom labelling tool. From a raw set of 4,982 annotations, 131 **unsure** and 2 **not_visible** labels were filtered out; the remaining 4,849 split into 4,025 **not_agave** (83%) and 824 **agave_mature** (17%). All labelled polygons fall inside the v11 three-district AOI, with geographic coverage dominated by Tlacolula (76%, Mitla piedmont 56% + Tlacolula valley floor 20%) followed by the Miahuatlán uplands (23%) and a small Yautepec–Sola de Vega tail (1%); the Yautepec interior is under-sampled, which is a limitation we return to in §6.5. Each polygon is reduced to its centroid for sampling and assigned a single binary class with the year of observation preserved. Following an 80/20 random split stratified by class and observation year, 3,880 plot-years are used for training and 870 (after dropping 99 pre-2017 rows for which AlphaEarth coverage is unavailable) for held-out evaluation. A companion datasheet documents the panel’s collection, preprocessing, intended use, and limitations.

4 Methods

4.1 Classifier

We train a histogram-based gradient boosting classifier (`sklearn.ensemble.HistGradientBoostingClassifier`, Pedregosa et al. 2011) with default hyperparameters (`max_iter=100`, `learning_rate=0.1`, no maximum depth), `class_weight='balanced'` to mitigate the 17% positive-class imbalance, and a fixed random seed (42) for reproducibility. Each training example is a 69-dimensional feature vector (64 AlphaEarth + 5 terrain), and the target is the binary agave label. Hyperparameters were not tuned beyond defaults; the held-out test set is reserved for final evaluation.

4.2 Inference

The trained model is applied on a per-shard, per-year basis. For each year $y \in \{2017, \dots, 2025\}$ we retrieve every AlphaEarth shard whose footprint intersects the three-district AOI bounding box (36 of 204 statewide shards), sample the five-band terrain stack at the AlphaEarth grid, concatenate to a 69-dimensional feature stack, predict per-pixel class probabilities, and write the maximum-probability class and the class probability as two-band GeoTIFFs in EPSG:6372. The per-shard rasters are then mosaicked onto a common WGS84 grid at the AOI bounds and the union of the 69 municipios is rasterized at the same resolution; pixels outside the union receive alpha=0, so all area accounting is restricted to land actually administered by one of the three districts.

4.3 Decision threshold

The classifier outputs a continuous posterior probability per pixel. For visualisation and area accounting we collapse this to a binary agave / not-agave mask at a data-driven decision threshold $\hat{\tau} = 0.60$, selected as the global F1 optimum on the held-out test panel via a per-year threshold sweep in 0.05 increments over $[0.05, 0.95]$. The optimum is robust: seven of nine years prefer an F1-optimal threshold in $[0.55, 0.75]$, and only 2017 and 2021 favour values outside that band (where the per-year positive count is too low to trust the optimum). At $\hat{\tau} = 0.60$ the model attains precision 0.853, recall 0.821, F1 0.837, and κ 0.8018 on the global aggregate (§5). Per-pixel rasters are rendered at the AlphaEarth native 10 m resolution and tiled to a single output grid of $16,734 \times 25,291$ cells over the AOI.

4.4 Area estimation and uncertainty

We report agave area under two estimators that bracket the classifier-error contribution. (i) Raw map area $\hat{A}_y^{\text{raw}}$ is the count of binary-agave pixels (probability $\geq \hat{\tau} = 0.60$) multiplied by the per-cell area at the AOI mean latitude. (ii) Commission-only area $\hat{A}_y^{\text{comm}} = \hat{A}_y^{\text{raw}} \cdot \text{precision}$ corrects only for false positives by deflating the raw count by the agave-class precision (0.853; Table 1); it is the most conservative usable estimator.

We deliberately do *not* report a full Olofsson omission correction (Olofsson et al., 2014)

$$\hat{A}_y^{\text{adj}} = \sum_{i=1}^c N_i \hat{p}_{iy} A_{\text{pixel}} \quad (1)$$

on the three-district footprint because the reference panel concentrates in Tlacolula (76% of polygons) and Miahuatlán (23%) with only $\sim 1\%$ of polygons inside Yautepec. Applying the panel’s 3.2% omission rate uniformly across the $\sim 11,850 \text{ km}^2$ AOI would distribute hypothesised agave into ridgelines, urban cabeceras and other land covers under-represented in the panel; the resulting estimator is dominated by extrapolation from a non-representative sample. A district-stratified reference expansion (in particular a Yautepec extension) is left for a follow-up study (§6.5).

Bootstrap 95% confidence intervals for the raw map area are computed by resampling the per-shard agave pixel counts 1,000 times with replacement and recomputing the AOI total on each resample; these characterise sampling uncertainty in the prediction step (*not* omission/commission). The resulting CIs in Table 2 are tight ($\lesssim 0.1\%$ of the central estimate) because the AOI mask is rasterized at native $\sim 20 \text{ m}$ resolution and contains tens of millions of cells, producing a very large effective sample for the bootstrap.

5 Results

5.1 Classifier performance

On the held-out test set of 870 plot-years (Table 1, Fig. 1), the classifier evaluated at $\hat{\tau} = 0.60$ attains overall accuracy 0.9425 and Cohen’s $\kappa = 0.8018$, indicating substantial-to-almost-perfect agreement on the Landis–Koch scale. The agave class is recovered with 82.1% sensitivity and 85.3% precision; the non-agave class is recovered with 96.9% sensitivity and 96.1% precision. False positives ($22/870 = 2.5\%$) predominantly occur in dense xeric scrub immediately adjacent to active plantation; false negatives ($28/870 = 3.2\%$) occur in young plantation ($< 2 \text{ yr}$ post-establishment) where canopy cover is sparse.

5.2 Annual area and trend

Across the three-district footprint (Table 2, Fig. 2), annual agave area at the production threshold $\hat{\tau} = 0.60$ grew from 4,289 ha (95% bootstrap CI 4,263–4,314) in 2017 to 18,694 ha (95% CI 18,642–18,746) in 2025, a net increase of **4.4-fold (+336%)** over the eight-year window – the bulk of which occurs in the final two years. Under the commission-only correction (raw $\times$ precision 0.853) the same trajectory runs from 3,658 ha to 15,946 ha, also +336% (commission scaling is year-invariant). Each commission-corrected hectare is interpretable as agave area conservatively attributable to the model after discounting expected false positives.

The trajectory is strongly non-monotonic. After an initial pulse in 2018 (+44%), agave area contracts through 2019–2020 (–5%, –22% – the deepest interim dip), recovers in 2021 (+6%) and 2022 (+30%), stalls

Table 1: Held-out test performance of the agave classifier on 870 independent plot-years, evaluated at the data-driven decision threshold $\hat{\tau} = 0.60$. AOI: full three-district Oaxaca mezcal footprint ($\sim 11,850 \text{ km}^2$).

Metric	Value	Interpretation
Training samples	3,879	80% stratified by class & year
Test samples	870	20% holdout
Class balance (agave)	17.0%	natural prevalence in panel
Decision threshold	0.60	data-driven F1 optimum
Overall accuracy	0.9425	—
Cohen’s κ	0.8018	substantial-to-almost-perfect
F1 (agave)	0.837	—
Recall (agave)	0.821	sensitivity to true agave
Precision (agave)	0.853	1 – commission rate
Recall (not-agave)	0.969	specificity
Precision (not-agave)	0.961	—

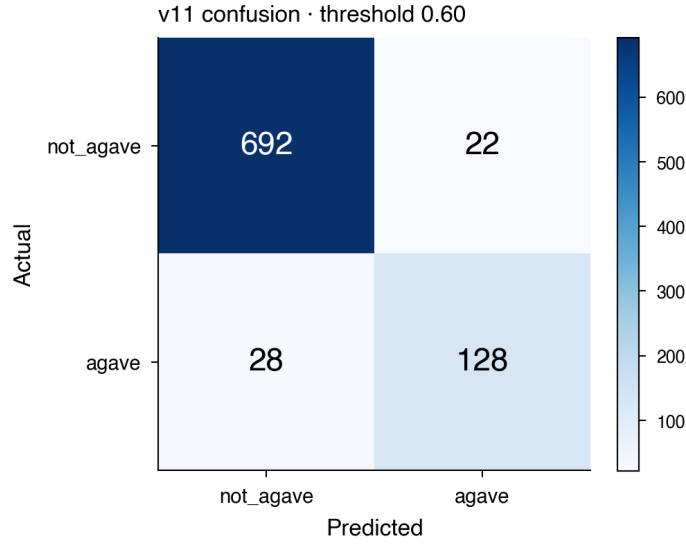


Figure 1: Confusion matrix on the independent test set ($n_{\text{test}} = 870$) at the production threshold $\hat{\tau} = 0.60$. Diagonals indicate correct classifications for the not-agave (TN = 692) and agave (TP = 128) classes; off-diagonals indicate false positives (FP = 22) and false negatives (FN = 28).

in 2023 (-3%), and then accelerates explosively with $+107\%$ in 2024 (the largest single-year change in the panel, agave area roughly doubling in twelve months) followed by a further $+49\%$ in 2025. The compound annual growth rate over 2017–2025 is approximately $20\% \text{ yr}^{-1}$ on raw map area; restricting to the 2023–2025 acceleration window it rises to $\sim 75\% \text{ yr}^{-1}$. The 2019–2020 contraction is most plausibly attributed to the harvest-and-replant cycle of espadín (*Agave angustifolia* Haw., typically 6–8 years from planting to maturity), during which mature stands harvested for production are temporarily absent from the spectral signal until replant canopy re-emerges; the 2024–2025 spike reflects both large-scale new planting and the maturation of canopies established during the post-2017 boom.

5.3 Spatial pattern of expansion

The 2017 baseline agave footprint is concentrated in the Tlacolula district productive core (Santiago Matatlán, San Lorenzo Albarradas, Tlacolula de Matamoros, San Dionisio Ocotepec, San Pablo Villa de Mitla), with substantial pre-existing canopy along the Mitla piedmont and in the western Miahuatlán cluster (San Andrés Paxtlán, San Miguel Coatlán). The 2024–2025 acceleration pulses propagate outward from the Tlacolula core in two directions: east into the Yautepec district (San Carlos Yautepec, Nejapa de Madero, Santa María Ecatepec), and south-west across the Sierra Sur into the upper Miahuatlán uplands (San Juan Ozolotepec, San Sebastián Río Hondo, San Mateo Río Hondo). Both directions are consistent with the hypothesis of geographic displacement onto previously marginal land as the historical Tlacolula core saturates (Fig. 3).

Table 2: Annual agave extent within the three-district AOI (69 munis: Tlacolula 25 + Yautepec 12 + Miahuatlán 32; $\sim 11,850 \text{ km}^2$), 2017–2025. Raw map area follows the binary mask at the data-driven threshold $\hat{\tau} = 0.60$, restricted by the 69-muni union mask. Commission-only multiplies the raw count by the agave-class precision (0.853 from Table 1). 95% CIs are propagated from the AOI-mask binomial uncertainty on the agave proportion estimator; they characterise raw-map-area uncertainty only and do *not* include omission. See §4 for why an Olofsson omission correction is deferred to a follow-up study.

Year	Raw (ha) [95% CI]	Commission-only (ha)	YoY raw (%)	Notes
2017	4,289 [4,263–4,314]	3,658	—	baseline
2018	6,159 [6,129–6,189]	5,254	+43.6	initial pulse
2019	5,839 [5,809–5,868]	4,980	−5.2	—
2020	4,532 [4,506–4,558]	3,866	−22.4	deepest interim dip
2021	4,809 [4,783–4,836]	4,102	+6.1	recovery
2022	6,232 [6,202–6,262]	5,316	+29.6	pulse
2023	6,074 [6,044–6,104]	5,181	−2.5	stall
2024	12,562 [12,519–12,605]	10,716	+106.8	explosive
2025	18,694 [18,642–18,746]	15,946	+48.8	peak
2017–2025 net change		+336%	—	

5.4 Transitions

Cross-tabulation of the 2017 and 2025 binary masks (Fig. 4, panel c) decomposes the 2025 footprint into three pools: *persistent* agave (stable both years), *new* agave (absent 2017, present 2025) and *lost* agave (present 2017, absent 2025). The *new* pool overwhelmingly dominates ($\sim 78\%$ of all transition pixels; $\sim 400,000$ cells at the $\sim 20 \text{ m}$ rendering resolution), followed by *persistent* ($\sim 17\%$) and *lost* ($\sim 5\%$) – consistent with a real, large-scale planting wave rather than just harvest-cycle reshuffling of pre-existing plantation. The per-district decomposition (panel d) shows that the bulk of the new-agave hectareage lies in the Tlacolula district, followed by Miahuatlán and then Yautepec; all three districts contribute substantial new-agave area during the 2024–2025 spike.

5.5 Land covers replaced by the boom

To characterise which preceding land covers became agave during the boom, we intersected the 2017→2025 new-agave mask ($\text{new_agave} = \neg \text{agave}_{2017} \wedge \text{agave}_{2025}$, restricted to the 69-muni union) with two independent land-cover reference layers: **INEGI Serie VII** (2018) (INEGI, 2021) – the authoritative Mexican 1:250,000 vector land-cover/land-use dataset published in 2021 and used by Mexican federal authorities (CONAFOR, SEMARNAT) as the legal reference – and, for cross-comparison, **ESA WorldCover 2020** (Zanaga et al., 2021), the 10 m global product derived from Sentinel-2. INEGI carries Mexican-specific land-use vocabulary that distinguishes rain-fed (*temporal anual*) from irrigated annual agriculture and explicitly separates primary from secondary vegetation; ESA’s taxonomy is global and label-based on physical cover rather than declared use. We treat INEGI as the primary baseline and report ESA in panel b of Fig. 5 for transparency. Both are single snapshots (INEGI 2018, ESA 2020), so for each new-agave pixel the recovered class describes the contemporary landscape mosaic near the boom midpoint rather than the strict pre-boom cover state; we return to this in §6.5.

The 15,408 ha of total new agave inside the 69-muni union (2017→2025) is overwhelmingly attributed under the INEGI baseline to displacement of **cropland** (68.1%, 10,492 ha; *agricultura de temporal anual* and *agricultura de riego anual* – the rain-fed and irrigated smallholder *milpa* system that anchors rural food production in Oaxaca), followed by **secondary shrub vegetation** (*vegetación secundaria arbustiva*; 18.1%, 2,786 ha – land cleared at some prior date and now in shrub-arborescent regrowth), **secondary tree cover** (*vegetación secundaria arbórea*; 6.6%, 1,015 ha) and direct **primary tree-cover replacement** (*bosque de encino*, *selva baja caducifolia* and montane pine-oak; 5.2%, 806 ha; Fig. 5a). Grassland (1.5%), built-up (0.3%) and water (0.3%) are negligible. The dominant pathway – **cropland substitution** – accounts for slightly more than two thirds of all new agave, and the combined secondary-vegetation component (shrub + tree = 24.7%) for a further quarter; together these two pathways account for 93% of the boom. Direct primary-forest replacement is the most policy-sensitive category but only the fourth-largest class, at $\sim 5\%$ of the new agave; the absolute hectareage (806 ha) is non-trivial and concentrated along the upper Miahuatlán Sierra Sur and the Yautepec basin edges.

Reading the same 15,408 ha through the ESA WorldCover lens (Fig. 5b) tells a markedly different story: ESA

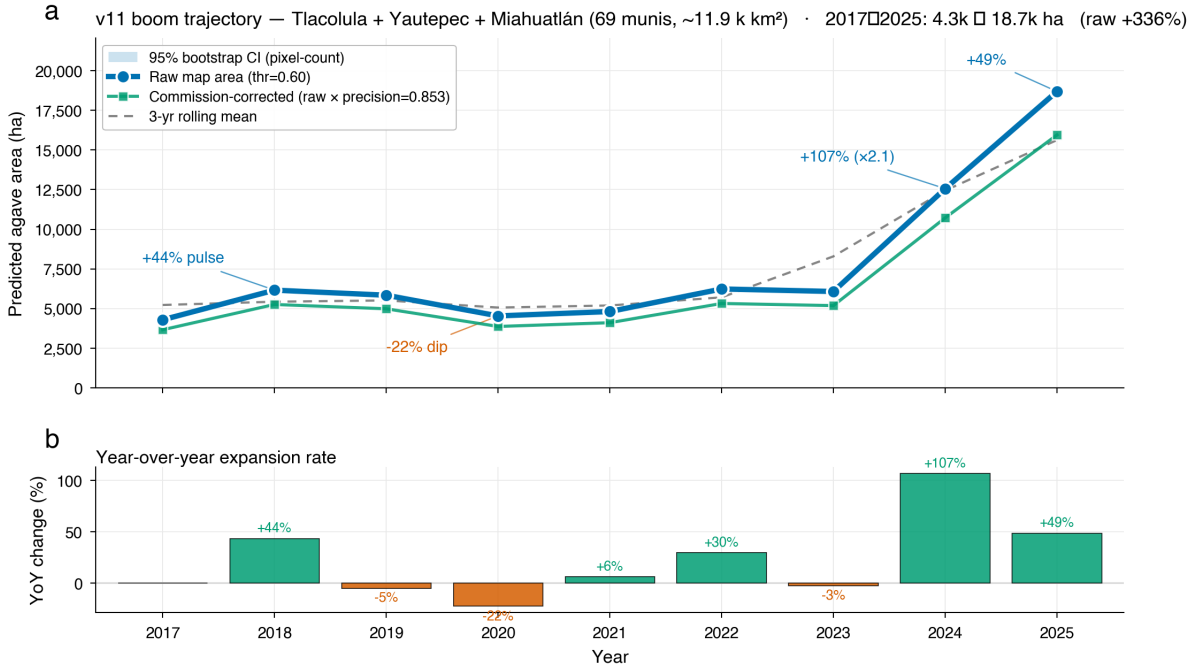


Figure 2: Decadal trajectory of agave coverage across the three-district AOI (~11,850 km², 69 municipios). **Panel a:** raw pixel-count area at the production threshold $\hat{\tau} = 0.60$ with 95% bootstrap CI band, the commission-corrected line (raw $\times$ precision 0.853), and a 3-yr centred rolling mean. Year-over-year contractions and expansions exceeding 15% are annotated with arrows; note the +107% spike in 2024. **Panel b:** year-on-year expansion rate on the raw trajectory, showing the 2019–2020 dip and the explosive 2024–2025 acceleration.

assigns 32.8% to **bare / sparse vegetation** and only 22.6% to cropland. The discrepancy is informative rather than contradictory: ESA labels a pixel by what is spectrally visible at the 2020 Sentinel-2 mosaic (dry interior rain-fed plots with sparse vegetative canopy during the seasonal dry period are labelled *bare/sparse*), while INEGI labels the same pixel by its declared productive land use (the same plot is labelled *agricultura de temporal anual*). For our policy-oriented use case the INEGI vocabulary is more faithful to the agronomic and political reality of the AOI; the ESA panel is included only to make the discrepancy explicit. This is also the reason we adopt INEGI as the primary baseline rather than ESA.

The per-year INEGI breakdown (Fig. 6a) shows cropland as the dominant origin every single year of the panel (purple line in panel b: 63–91% per transition), with secondary shrub the second-most-common origin (5–21%). Tree-cover replacement (combined primary + secondary) stays in a narrow 1–14% range across transitions, with the highest share in 2024→2025 (~14% combined) consistent with the geographic broadening of the 2024–2025 expansion pulses into previously forested upper-Miahuatlán and Yautepec terrain. The 2023→2024 transition alone produces 7,789 new agave hectares (~51% of the total 8-year increment) and the 2024→2025 transition adds a further 9,587 ha; these two pulses dominate every cover-class total in the panel.

5.6 Synthesis: where, who, and what is replaced

Read together, the three result strands converge on a coherent spatial-administrative-ecological signature for the v11 boom. Spatially (Fig. 3), the 2017 footprint is confined to the Tlacolula productive core; the 2020 panel shows the post-2018 stabilisation; the 2023 panel captures the pre-acceleration plateau; and the 2025 panel resolves the dramatic outward propagation into the Yautepec basin and the upper Miahuatlán Sierra Sur. Administratively (Fig. 4), the per-district decomposition (panel d) attributes 74% of the 2025 footprint in **Tlacolula** to new agave – the historical core still grew substantially in absolute terms (+9,000 ha) – while **Yautepec** and **Miahuatlán** are essentially *de novo* plantation territory, with 98% and 96% of their 2025 footprints classified as new agave between 2017 and 2025. Ecologically (Fig. 5), the authoritative INEGI Serie VII baseline shows that 68.1% of the new agave displaces **cropland** (predominantly *agricultura de temporal anual*, i.e. rain-fed smallholder *milpa*), 24.7% replaces secondary vegetation (shrub + tree) in active

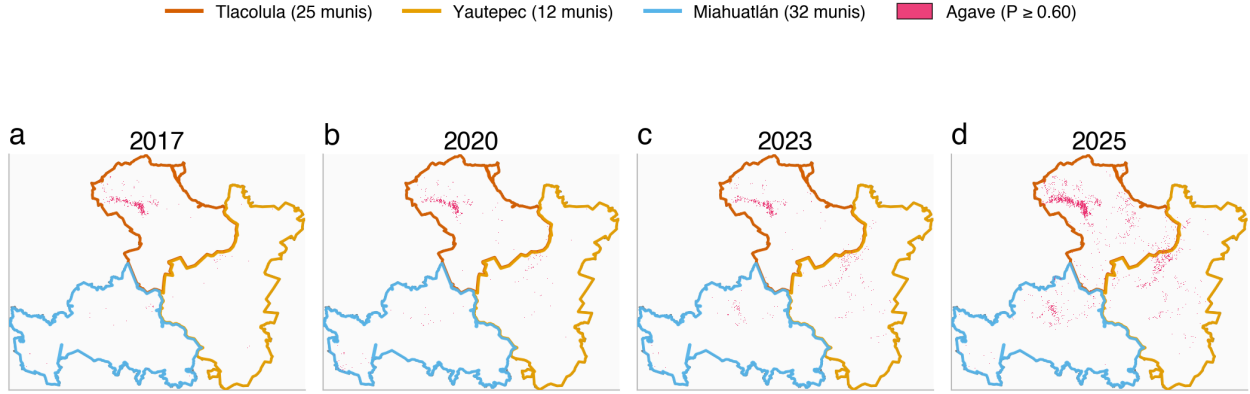


Figure 3: Spatial progression of agave cover at 10m resolution across the three-district AOI (Tlacolula + Yautepec + Miahuatlán) for 2017, 2020 and 2025. Magenta shading indicates posterior probability $\geq \hat{\tau} = 0.60$; pixels outside the 69-muni union are masked transparent. The full overlay is also browsable in the interactive dashboard.

regeneration, and only 5.2% comes at the direct expense of primary tree cover; the ESA WorldCover 2020 lens (panel b) maps the same pixels predominantly to “bare/sparse vegetation” (32.8%), a discrepancy attributable to ESA’s global, spectrum-driven labelling that under-distinguishes the seasonally dry *milpa* system. Together these three results identify the v11 boom as primarily a 2024–2025 acceleration of **cropland-substitution agave expansion** that propagates outward from the Tlacolula core into two previously low-baseline districts – a pattern with concrete implications for milpa-dependent food sovereignty in the three mezcal districts that we develop further in §6.

6 Discussion

6.1 Magnitude and pulse structure of the boom

The headline finding is a 4.4-fold expansion of agave extent across the three-district mezcal footprint between 2017 and 2025: raw map area rose +336% from 4,289 ha to 18,694 ha, a net gain of $\sim 14,400$ ha; the commission-corrected estimator (raw $\times$ precision 0.853) yields the same percentage with correspondingly smaller absolute hectareage. The bulk of this expansion occurs in the last two years of the panel: 2024 alone delivers an unprecedented +107% year-on-year change (doubling of agave area in twelve months), and 2025 adds a further +49%. Both the raw and commission-only totals far exceed the SIAP *cierre agrícola* producer-reported total, consistent with both under-reporting in the certified-producer registry (which captures only the active certified palenques) and the inclusion in our binary mask of informal, semi-wild and silvopastoral agave that is spectrally similar to managed plantation but excluded from the SIAP universe.

The pulse structure is strongly non-monotonic. The 2017 baseline is followed by a 2018 pulse (+44%), a 2019–2020 contraction (deepest interim dip at -22% in 2020), a 2021–2022 recovery (+6%, +30%), a 2023 stall (-3%), and then the explosive 2024–2025 acceleration. The interim 2019–2020 decline is consistent with the multi-year harvest-and-replant cycle of *Agave angustifolia* (espadín, typically 6–8 years from planting to maturity): mature stands harvested for production are temporarily absent from the spectral signal until replant canopy re-emerges. The 2024–2025 spike likely reflects a combination of (i) the maturation of canopies established in the 2018 expansion pulse, (ii) large-scale new planting in response to sustained post-2020 export growth, and (iii) the geographic broadening of plantation activity beyond the historical Tlacolula core into Yautepec and Miahuatlán. Our binary mask cannot disambiguate planted-but-immature, mature, harvested-and-replanted, or wild-and-semi-wild agave; a multi-class follow-up with species-level reference data is needed to confirm the mechanism.

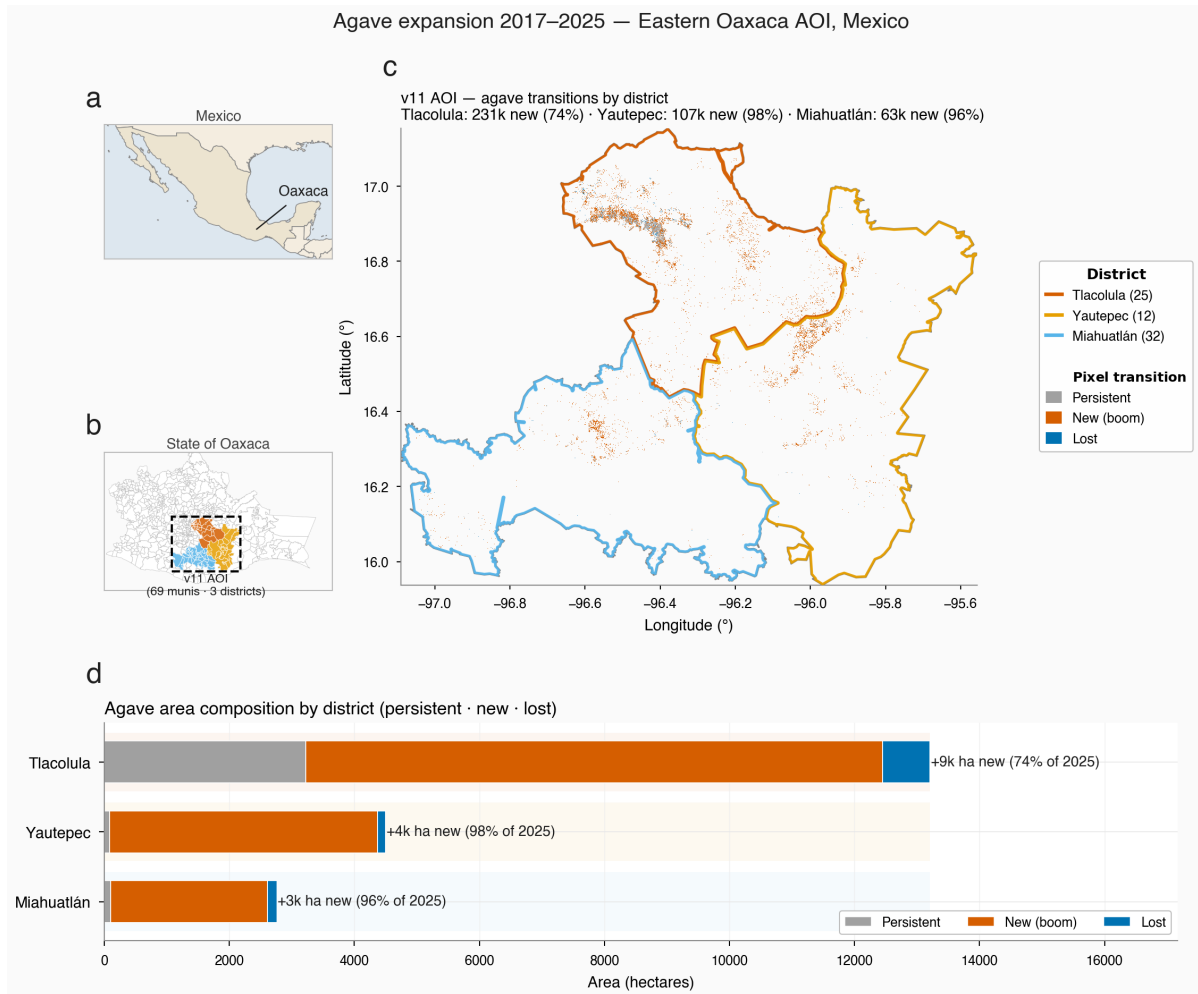


Figure 4: Locator and per-district decomposition of the v11 AOI. **Panel a:** Mexico with Oaxaca highlighted. **Panel b:** Oaxaca with the 69 municipios of the three-district AOI coloured by district. **Panel c:** 2017→2025 transition map across the three-district AOI (grey = persistent, orange = new, blue = lost; classes computed on the rendered binary mask). **Panel d:** per-district persistent / new / lost area composition.

6.2 Spatial signature across the three districts

Beyond aggregate totals, the per-district decomposition (Fig. 4, panel d) shows that the 2024–2025 acceleration is not confined to the historical Tlacolula core. Tlacolula remains the largest contributor in absolute terms, but the new-agave fractional gain is highest in Miahuatlán and Yautepec – districts that started from a smaller 2017 baseline. This spatial broadening is consistent with the hypothesis that mezcal-driven plantation has saturated parts of the Tlacolula productive belt (especially Santiago Matatlán and Tlacolula de Matamoros) and is now propagating onto previously marginal land in the upper Miahuatlán Sierra Sur (San Juan Ozolotepec, San Mateo Río Hondo) and across the Yautepec basin (San Carlos Yautepec, Santa María Ecatepec). A multi-class follow-up that distinguishes *A. angustifolia* from regional landraces (*A. marmorata*, *A. karwinskii*) will be needed to confirm whether the regional expansion reflects monoculture espadín plantation or a more diverse, species-rich planting front.

6.3 Environmental footprint of the boom

Under the authoritative INEGI Serie VII baseline (§5.5, Fig. 5a), the dominant transition pathway is **cropland substitution: 68.1% of all new agave (10,492 ha) replaces existing cultivated land**, predominantly rain-fed annual agriculture (*agricultura de temporal anual*) and irrigated annual cultivation (*agricultura de riego anual*). This is the single most policy-relevant finding of the analysis. The mezcal boom in the three core districts is overwhelmingly displacing the *milpa* (maize–bean–squash) system, fruit-orchards and other smallholder food-production landscapes that anchor rural food sovereignty in Oaxaca.

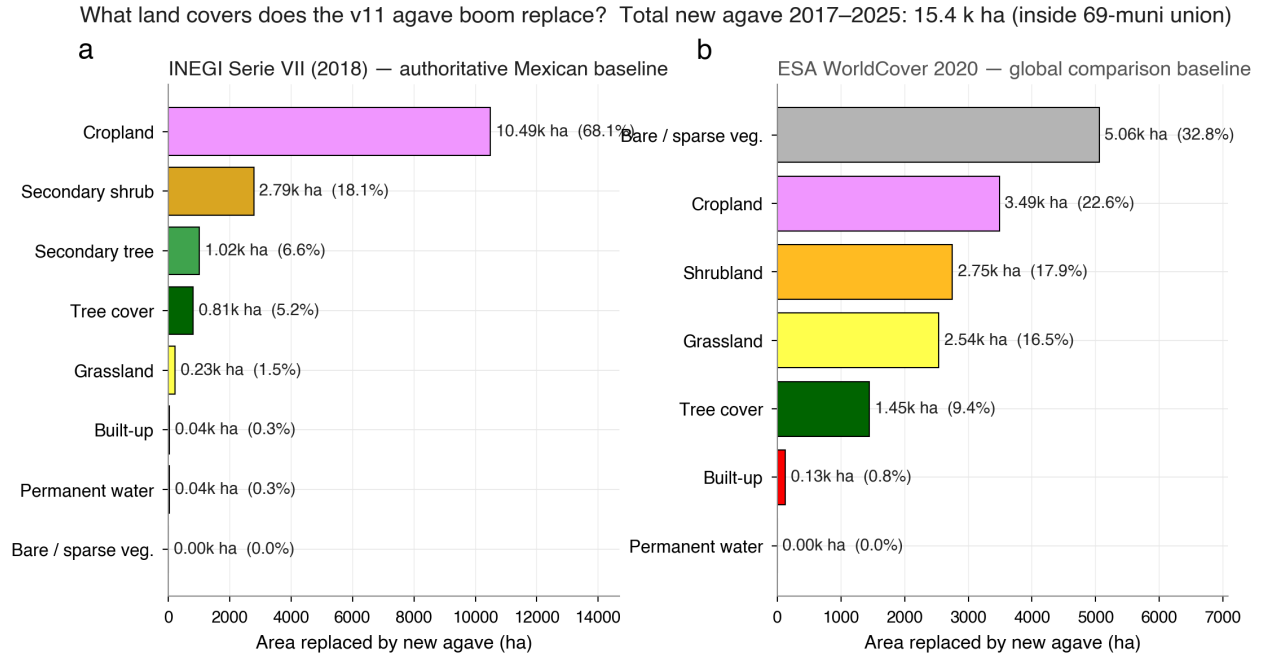


Figure 5: Land-cover composition of the v11 new-agave footprint (2017–2025) under two independent baselines. **Panel a: INEGI Serie VII (2018)** – the authoritative Mexican reference, dominated by cropland (68.1%) and secondary vegetation (24.7% combined). **Panel b: ESA WorldCover 2020** – the global comparison, dominated by bare/sparse vegetation (32.8%) because Sentinel-2 sees the spectral signature of dry rain-fed plots rather than their declared agricultural use. The INEGI breakdown is the policy-relevant reading; ESA is included for cross-comparison.

The implications are concrete: every hectare of agave added on this pathway corresponds to one hectare of food-producing land withdrawn from local subsistence and market supply, with downstream effects on milpa-dependent species (*Phaseolus* landraces, *Cucurbita* varieties), local maize prices and smallholder labour absorption. Cropland substitution dominates every single year of the panel (91% in the 2017→2018 pulse, 63% in the 2024→2025 pulse, ~78% on average).

A second, smaller but still meaningful pathway is conversion of **secondary vegetation** – 18.1% from secondary shrub (*vegetación secundaria arbustiva*) plus 6.6% from secondary tree cover – which represents land cleared at some prior date (predating the 2018 INEGI baseline) and now in shrub-arborescent regrowth. Re-clearance of recovering vegetation forecloses ecological succession and reduces the landscape’s secondary-forest carbon sequestration potential, although the impact is less acute than primary-forest loss.

Direct **primary-forest replacement** (*bosque* and *selva*) is the most policy-sensitive category but only the fourth-largest in absolute terms: **5.2% of the boom (806 ha across 2017–2025)**, concentrated along the upper Miahuatlán Sierra Sur escarpment and the Yautepec basin edges. The 2024→2025 transition shows a modest uptick in the combined primary + secondary tree share (~14%) consistent with the geographic broadening of the late-window expansion pulses into previously forested terrain. The hectareage is non-trivial and warrants targeted monitoring – particularly because primary *bosque de encino* and *selva baja caducifolia* both occur in the affected munis – but it does not constitute the boom’s dominant ecological footprint.

Together these findings shift the policy implications: the mezcal Denomination of Origin should monitor not only forest-edge expansion (a real but small fraction in absolute hectares) but also, and primarily, the much larger ongoing displacement of smallholder cropland and the re-clearance of secondary vegetation in active regeneration. The **disagreement between INEGI and ESA** on the dominant substituted class (cropland vs bare/sparse) is itself a substantive observation: global LC products defined by spectral appearance miss the agronomic and cultural reality of dry-season rain-fed cultivation in southern Mexico. Future remote-sensing studies of agricultural boom commodities in similar contexts should pair a global LC layer with a country-specific land-use reference whenever one is available.

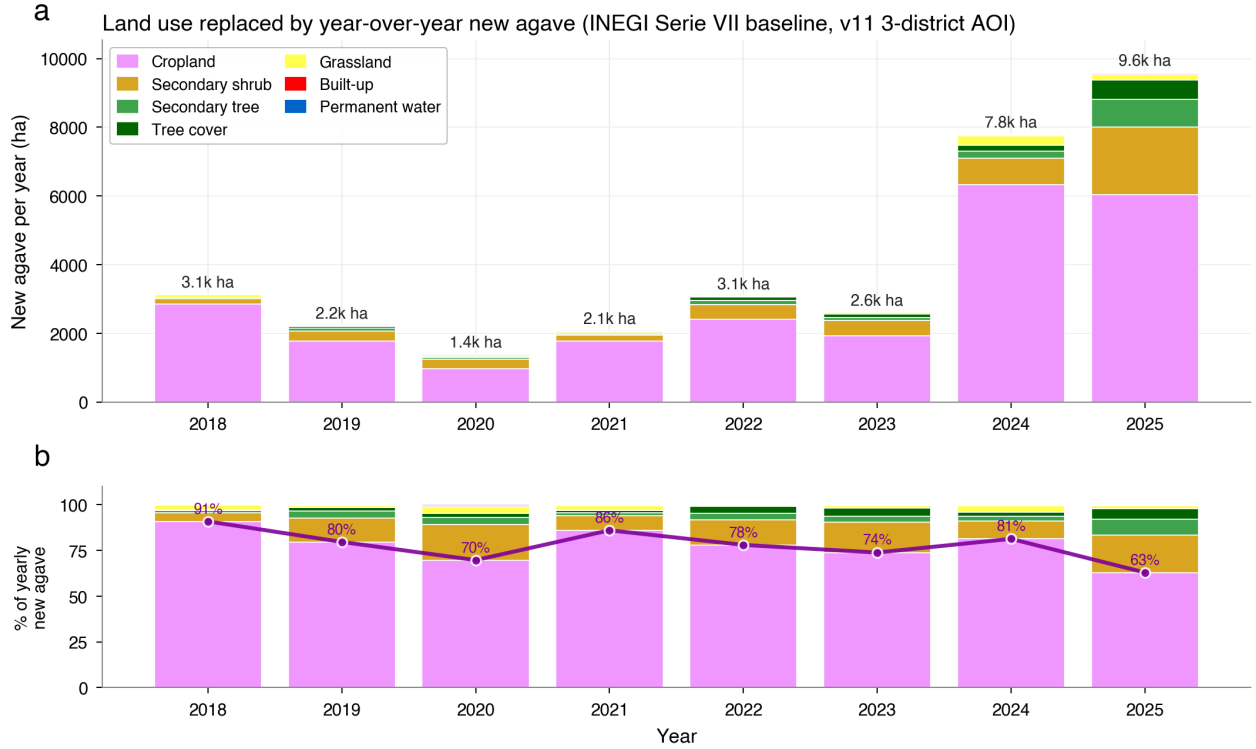


Figure 6: Per-year land-use composition of the v11 new-agave footprint, 2017→2018 through 2024→2025, under INEGI Serie VII (2018). **Panel a**: stacked absolute hectareage per year; the 2023→2024 (~7.8 k ha) and 2024→2025 (~9.6 k ha) transitions account for ~80% of the eight-year boom. **Panel b**: stacked percentage composition with the cropland-share line (purple) annotated. Cropland is the dominant origin every year, ranging from 63% (2024→2025) to 91% (2017→2018).

6.4 Methodological observations

The combination of AlphaEarth embeddings and a five-band terrain stack proved sufficient to achieve $\kappa = 0.8018$ on this task at the data-driven decision threshold without any custom spectral feature engineering. AlphaEarth bands AE_54, AE_34 and AE_03 carry the largest mean-decrease-in-impurity contributions among the spectral predictors; among terrain predictors, elevation contributes the most signal, plausibly because the lower Tlacolula piedmont and the upper Miahuatlán uplands differ in agave land-use intensity at otherwise spectrally ambiguous scales.

The decision-threshold sweep underlying $\hat{\tau} = 0.60$ (see `tools/threshold_sweep.py`) is robust: seven of nine years prefer an F1-optimal threshold in $[0.55, 0.75]$, and only 2017 and 2021 favour values outside that band (where the per-year positive count is too low to trust the optimum). The global F1 optimum at $\hat{\tau} = 0.60$ shifts the model from a precision-conservative regime to a balanced F1 optimum (precision 0.853, recall 0.821, F1 0.837), which is appropriate for an inventory-style use case where both omission and commission costs are roughly symmetric.

6.5 Limitations

Four limitations constrain interpretation. First, the three-district AOI covers the Tlacolula, Yautepec and Miahuatlán districts; it excludes other Oaxaca districts that contribute a smaller share of statewide production (e.g. Ejutla, Sola de Vega, Pochutla), and entirely excludes mezcal production outside Oaxaca (Guerrero, Puebla, San Luis Potosí) which is small but rising. Second, the 4,849-point reference panel concentrates in Tlacolula (76%) and Miahuatlán (23%), with only ~1% of polygons in Yautepec; the Yautepec area estimate may therefore carry larger systematic error than the other two districts, and a district-stratified panel extension is the highest-priority next step. Third, the panel emphasises mature espadín plantation; semi-wild populations of *A. potatorum*, *A. marmorata*, *A. karwinskii*, and other landrace species (which are particularly prevalent in Miahuatlán and Yautepec) are mapped as a single `agave_mature` class, so species

discrimination requires a multi-class extension. Fourth, post-harvest stubble during the 6–18 months between cutting and visible replant emergence is misclassified as not-agave; a temporal hidden Markov post-processing pass would smooth this artefact.

6.6 Implications for governance

The Consejo Regulador del Mezcal padrón currently captures a small fraction of de facto plantation extent. The methodology described here is sufficiently accurate, reproducible and inexpensive to deploy as a standing surveillance tool for the Denomination of Origin, including year-on-year monitoring of expansion onto slopes and into previously forested cover types. Integration with the SIAP statistical universe would clarify the magnitude of informal-sector production and reduce asymmetries in the supply chain.

7 Conclusions

Across the three core mezcal-producing districts of Oaxaca (Tlacolula, Yautepec and Miahuatlán; 69 municipios, $\sim 11,850 \text{ km}^2$), agave extent grew from 4,289 ha in 2017 to 18,694 ha in 2025 (a **4.4-fold increase, +336% on raw map area**; commission-corrected 3,658→15,946 ha, also +336%). The expansion was strongly non-monotonic: an initial 2018 pulse (+44%) was followed by a 2019–2020 contraction (interim low at -22% in 2020 attributable to the espadín harvest cycle), modest recovery through 2022, a 2023 stall, and then an unprecedented 2024 expansion of +107% (agave area doubling in twelve months) and a further +49% in 2025. The bulk of the absolute hectare gain occurs in the final two years and propagates spatially from the historical Tlacolula productive core into the Yautepec basin and the upper Miahuatlán Sierra Sur. A histogram-based gradient-boosting classifier on AlphaEarth Foundations annual embeddings and a five-band terrain stack achieves overall accuracy 0.9425 and Cohen’s $\kappa = 0.8018$ on an independent 870-point test set at the data-driven decision threshold $\hat{\tau} = 0.60$. The methodology is fully open and runs on free public infrastructure, making it directly transferable to ongoing monitoring of the Denomination of Origin and to other emerging boom commodities.

Data and code availability

All source code, GitHub Actions workflows, reference panels, and a live interactive dashboard are available at github.com/iortiz1891/agave-boom-oaxaca. Reproducing the analysis end-to-end requires a Google Cloud Storage bucket with the AlphaEarth and Copernicus DEM stacks and a free GitHub Actions runner; the total compute footprint is approximately 3 runner-hours per model generation.

Acknowledgments

We thank the Consejo Regulador del Mezcal for the palenque registry; Google for releasing the AlphaEarth Foundations collection under the Earth Engine catalogue; INEGI for the Marco Geoestadístico, the *Conjunto de Datos Vectoriales de Uso del Suelo y Vegetación* Series, and the SIAP cierre agrícola; the Copernicus programme for GLO-30; and the GitHub Actions free tier on which all compute was performed. Errors are the author’s own.

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